

### 5-3c Does Drug Interdiction Increase or Decrease Drug-Related Crime?

A persistent problem facing our society is the use of illegal drugs, such as heroin, cocaine, ecstasy, and methamphetamine. Drug use has several adverse effects. One is that drug dependence can ruin the lives of drug users and their families. Another is that drug addicts often turn to robbery and other violent crimes to obtain the money needed to support their habit. To discourage the use of illegal drugs, the U.S. government devotes billions of dollars each year to reducing the flow of drugs into the country. Let's use the tools of supply and demand to examine this policy of drug interdiction.

Suppose the government increases the number of federal agents devoted to the war on drugs. What happens in the market for illegal drugs? As usual, we answer this question in three steps. First, we consider whether the supply or demand curve shifts. Second, we consider the direction of the shift. Third, we see how the shift affects the equilibrium price and quantity.

Although the purpose of drug interdiction is to reduce drug use, its direct impact is on the sellers of drugs rather than on the buyers. When the government stops some drugs from entering the country and arrests more smugglers, it raises the cost of selling drugs and, therefore, reduces the quantity of drugs supplied at any given price. The demand for drugs—the amount buyers want at any given price—remains the same. As panel (a) of [Figure 9](#) shows, interdiction shifts the supply curve to the left from  $S_1$  to  $S_2$  without changing the demand curve. The equilibrium price of drugs rises from  $P_1$  to  $P_2$ , and the equilibrium quantity falls from  $Q_1$  to  $Q_2$ . The fall in the equilibrium quantity shows that drug interdiction does reduce drug use.

#### Figure 9

#### Policies to Reduce the Use of Illegal Drugs

Drug interdiction reduces the supply of drugs from  $S_1$  to  $S_2$ , as in panel (a). If the demand for drugs is inelastic, then the total amount paid by drug users rises, even as the amount of drug use falls. By contrast, drug education reduces the demand for drugs from  $D_1$  to  $D_2$ , as in panel (b). Because both price and quantity fall, the amount paid by drug users falls.

Two graphs compare (a) drug interdiction and (b) drug education in terms of their effect on the supply and demand for drugs. Graph (a), titled Drug Interdiction, plotting quantity of drugs on the horizontal axis and price of drugs on the vertical axis, shows a downward sloping demand curve intersected by two parallel upward sloping supply curves,  $S_1$  and  $S_2$ , with  $S_2$  closer to the origin than  $S_1$ . From where the demand curve intersects  $S_1$ , perpendiculars are dropped to the vertical axis and horizontal axis at  $P_1$  and  $Q_1$ , respectively. Similarly from where the demand curve intersects  $S_2$ , perpendiculars are dropped at  $P_2$  and  $Q_2$ . Three callouts pointing at the shifts from  $S_1$  to  $S_2$ , from  $P_1$  to  $P_2$ , and from  $Q_1$  to  $Q_2$ , state, "Drug interdiction reduces the supply of drugs ... which raises the price ... and reduces the quantity sold." Graph (b), titled Drug Education, plotting quantity of drugs on the horizontal axis and price of drugs on the vertical axis, shows an upward sloping supply curve intersected by two parallel downward sloping demand curves,  $D_1$  and  $D_2$ , with  $D_2$  closer to the vertical axis than  $D_1$ . From where the supply curve intersects  $D_1$ , perpendiculars are dropped to the vertical axis and horizontal axis at  $P_1$  and  $Q_1$ , respectively. Similarly from where the supply curve intersects  $D_2$ , perpendiculars are dropped at  $P_2$  and  $Q_2$ . Three callouts pointing at the shifts from  $D_1$  to  $D_2$ , from  $P_1$  to  $P_2$ , and from  $Q_1$  to  $Q_2$ , state, "Drug education reduces the demand for drugs ... which reduces the price ... and reduces the quantity sold."

But what about the amount of drug-related crime? To answer this question, consider the total amount that drug users pay for the drugs they buy. Because few drug addicts are likely to break their destructive habits in response to a higher price, it is likely that the demand for drugs is inelastic, as it is drawn in the figure. If demand is inelastic, then an increase in price raises total revenue in the drug market. That is, because drug interdiction raises the price of drugs proportionately more than it reduces drug use, it raises the total amount of money that drug users pay for drugs. Addicts who already had to steal to support their habits would now have an even greater need for quick cash. Thus, drug interdiction could increase drug-related crime.

Because of this adverse effect of drug interdiction, some analysts argue for alternative approaches to the drug problem. Rather than trying to reduce the supply of drugs, policymakers might try to reduce the demand by pursuing a policy of drug education. Successful drug education has the effects shown in panel (b) of [Figure 9](#). The demand curve shifts to the left from  $D_1$  to  $D_2$ . As a result, the equilibrium quantity falls from  $Q_1$  to  $Q_2$ , and the equilibrium price falls from  $P_1$  to  $P_2$ . Total revenue,  $P \times Q$ , also falls. Thus, in contrast to drug interdiction, drug education can reduce both drug use and drug-related crime.

Advocates of drug interdiction might argue that the long-run effects of this policy are different from the short-run effects because the elasticity of demand depends on the time horizon. The demand for drugs is probably inelastic over short periods because higher prices do not substantially affect drug use by established addicts. But it may be more elastic over longer periods because higher prices would discourage experimentation with drugs among the young and, over time, lead to fewer drug addicts. In this case, drug interdiction would increase drug-related crime in the short run but decrease it in the long run.

## QuickQuiz

8. An increase in the supply of grain will reduce the total revenue grain producers receive if
- a. the supply curve is inelastic.
  - b. the supply curve is elastic.
  - c. the demand curve is inelastic.
  - d. the demand curve is elastic.
9. In competitive markets, farmers adopt new technologies that will eventually reduce their revenue because
- a. each farmer is a price taker.
  - b. farmers are short-sighted.
  - c. regulation requires the use of best practices.
  - d. consumers pressure farmers to lower prices.
10. Because the demand curve for oil is \_\_\_\_\_ elastic in the long run, OPEC's reduction in the supply of oil had a \_\_\_\_\_ impact on the price in the long run than it did in the short run.
- a. less; smaller
  - b. less; larger
  - c. more; smaller
  - d. more; larger
11. Over time, technological advances increase consumers' incomes and reduce the price of smartphones. Each of these forces increases the amount consumers spend on smartphones if the income elasticity of demand is greater than \_\_\_\_\_ and the price elasticity of demand is greater than \_\_\_\_\_.
- a. zero; zero
  - b. zero; one

c. one; zero

d. one; one

Chapter 5: Elasticity and Its Application: 5-3c Does Drug Interdiction Increase or Decrease Drug-Related Crime?

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